

Consistency Study of Alpha Vantage and SEC Financial Data

Scope: 7 companies · FY2019–FY2024 · 7 common metrics · 294 eligible cells

1. Scope of This Study

This study covers the team’s original concerns, plus an additional test on a more complex derived metric:

1. Whether seven common financial metrics keep the same meaning across years within the same company and across peers;
2. Whether Alpha Vantage’s current historical values are closer to the SEC’s **as-first-reported** values or to **as-restated** values;
3. The frequency, size, concentration, and severity of data gaps;
4. Whether metrics with the same name are truly comparable within the same SEC SIC industry;
5. Whether Alpha Vantage’s basic financial endpoints cover industry-specific or company-specific KPIs;
6. (Additional test) Whether a complex derived metric such as total debt can be handled by a single fixed formula.

This report uses seven companies, FY2019–FY2024, and seven common financial metrics, forming **294 predefined research cells (eligible cells)**. Among these, **290** have a complete SEC reference matching the product definition. The study also produced **720** SEC historical-version records, **126** peer-comparison records, and **13** special-KPI exploration records.

2. Research Objects, Metric Definitions, and Data Methodology

2.1 Sample and Peer Groups

Companies studied: MSFT, ORCL, XOM, CVX, F, GM, and IBM.

Study period: FY2019–FY2024. Each company uses its own fiscal year end. For example, Microsoft and Oracle have different fiscal year-end dates, so peer comparison must be interpreted using **both the fiscal-year label and the exact period-end date**.

SEC four-digit SIC peer groups:

Industry	Companies	SIC
Software	MSFT / ORCL	7372
Petroleum refining	XOM / CVX	2911
Motor vehicles	F / GM	3711
Computer and office equipment	IBM	3570

IBM is used separately for historical restatement, within-company definitional drift, and special-KPI cases.

2.2 Detailed Definitions of the Seven Common Metrics

Metric	Product definition used in this study	Main SEC concept	Period type
Total assets	Carrying amount of all assets recognized by the consolidated entity at period end	Assets	Instant, period-end balance
Total liabilities	Carrying amount of all liabilities recognized by the consolidated entity at period end	Liabilities ; if missing, may be derived within the same accession as assets minus equity including NCI	Instant
Shareholders' equity	Equity attributable to parent-company shareholders, excluding NCI	StockholdersEquity	Instant
Cash and cash equivalents	Unrestricted cash, demand deposits, and highly liquid near-maturity cash equivalents; excludes restricted cash	CashAndCashEquivalentsAtCarryingValue	Instant
Revenue	Broad consolidated top-line revenue: the highest-level income-statement revenue line covering all of the company's revenue activities	Usually Revenues , but verified company-year by company-year against the primary income statement	Duration, full fiscal year
Net income	Net income attributable to the parent company — not income available to common shareholders after preferred-stock or similar items	NetIncomeLoss	Duration

	Revenue		
Operating cash flow	Net cash provided by or used in operating activities for the full fiscal year	NetCashProvidedByUsedInOperatingActivities	Duration

2.2.1 What Restricted Cash Is, and Why This Study Excludes It

CashAndCashEquivalentsAtCarryingValue includes cash on hand, demand deposits, and short-term highly liquid investments that are readily convertible into known amounts of cash and subject to insignificant interest-rate risk.

Restricted cash is still cash owned by the company, but its use is limited by contract, law, regulation, collateral arrangements, or similar restrictions.

Cash-flow statements often tie out beginning and ending balances of “cash, cash equivalents, and restricted cash” together. That does **not** mean the ordinary balance-sheet cash line should automatically include restricted cash. For this study, the product definition of cash excludes restricted cash.

2.2.2 What “Broad Consolidated Revenue” Means

The Revenue metric in this study does not mean “any number with the word *revenue* in its label.” It means:

The highest-level total-revenue line on the consolidated income statement for the full fiscal year, covering all revenue sources that the company presents as revenue.

In the SEC taxonomy, Revenues is a broad concept. RevenueFromContractWithCustomerExcludingAssessedTax is narrower: it focuses on revenue from contracts with customers, excluding taxes collected on behalf of governments.

For oil companies, the narrower customer-revenue concept may be close to sales or customer revenue, but it may exclude other revenue components included in the company’s broad revenue line.

This study uses broad consolidated revenue for four reasons:

1. Alpha Vantage’s field is called totalRevenue, so ordinary users naturally interpret it as total company revenue, not merely one type of sales revenue;
2. The broad top-line concept is more suitable as a common “company total revenue” metric across industries;
3. It reveals whether a data vendor silently switches to a narrower sales line in certain years;
4. It ties out to the highest-level consolidated income-statement line, rather than treating one component of revenue as the whole.

A narrow sales or customer-revenue metric is not wrong in itself. If a product wants to show “sales excluding other revenue,” it can do so. The key requirement is that it must be **separately named** and applied **consistently** across companies and years. This study tests whether Alpha Vantage’s totalRevenue stably satisfies the pre-declared broad-revenue contract.

2.3 SEC Data Sources and How Their JSON Structures Are Used

2.3.1 Submissions API

The SEC company submissions JSON can be understood as a company’s **filing directory and metadata index**. Its core contents include:

- CIK, company name, ticker, SIC, and SIC description;
- Arrays of recent filings, including fields such as:
 - accessionNumber
 - filingDate
 - reportDate
 - form
 - primaryDocument
- References to additional JSON files for older filings.

This study uses the Submissions API to:

1. Confirm the company’s SEC SIC;
2. Identify target 10-K and 10-K/A filings;
3. Locate the filing accession number and primary HTML document;
4. Determine whether a later annual report exists after a given historical period.

The Submissions API does not directly provide all financial values. Its main role is to tell us which filings exist, when they were filed, and which documents they correspond to.

2.3.2 Company Facts API

The SEC Company Facts JSON is a structured aggregation of facts extracted from XBRL filings. Its structure is roughly:

```

facts
├── taxonomy namespace, such as us-gaap
│   └── concept, such as Assets or Revenues
│       ├── label
│       ├── description
│       └── units
│           ├── USD / shares / USD-per-share, etc.
│           └── fact records

```

Each fact record commonly contains:

- `val` : value;
- `start` / `end` : period start and end; instant facts usually only have `end` ;
- `accn` : the accession number of the filing the fact comes from;
- `filed` : filing date;
- `form` : 10-K, 10-Q, etc.;
- `fy` / `fp` : fiscal year and fiscal period marked by the filing;
- `frame` : SEC's standardized time frame, when available.

This study uses Company Facts to build:

- A version timeline for the same company–metric–fiscal-year across multiple filings;
- Values for exact standard concepts;
- Candidate matches between related concepts for the same period, such as parent equity versus equity including NCI.

Company Facts is not a complete substitute for the original filing. Its coverage of custom tags, dimensional facts, table context, and narrative KPIs is limited. Therefore, revenue scope, special KPIs, and complex debt still require going back to the filing HTML or Inline XBRL.

2.3.3 10-K HTML and Inline XBRL

Inline XBRL keeps two layers in the same filing document:

- Human-readable financial statements, tables, and text;
- Machine-readable concepts, contexts, units, and values.

This study uses filing HTML / Inline XBRL to:

- Confirm visible income-statement line names;
- Determine whether a revenue number is broad total revenue or narrow sales;
- Inspect tables or narrative sections containing industry KPIs;
- Preserve the actual source filing rather than linking only to a generic company page.

2.3.4 Alpha Vantage Data

The study checks four Alpha Vantage fundamentals endpoints:

- `BALANCE_SHEET`
- `INCOME_STATEMENT`
- `CASH_FLOW`
- `OVERVIEW`

The Alpha Vantage data used here is the **current historical snapshot at the time of collection**. This means the study cannot reconstruct what Alpha Vantage returned on the original filing date in 2020. It can only answer: *what does Alpha Vantage currently return for FY2020?*

2.4 Key Terms

XBRL Concept / Tag

An XBRL concept is a standardized semantic identifier, such as `Assets`. In plain language, a tag is a machine-readable label attached to a visible number in a filing. A number may be called “cash” in the filing, but that does not automatically mean its tag is the cash concept required by this study. We still need to inspect the tag, context, and scope.

Direct Tag

A direct tag means the filing directly reports the target concept as a fact. For example, if the balance sheet directly contains `Liabilities = 131.7B`, that is a direct `Liabilities` fact.

This is different from a derived value. If a filing does not contain a direct `Liabilities` fact, but the same accession contains:

```
Assets - StockholdersEquityIncludingPortionAttributableToNoncontrollingInterest
```

then liabilities can be derived using the balance-sheet identity. The derived result may be numerically correct, but its provenance must be shown as “derived.” It must not be described as a directly disclosed SEC line.

NCI and Equity Including NCI

NCI means noncontrolling interest. It refers to the portion of a consolidated subsidiary’s equity that is owned by shareholders other than the parent company. The subsidiary’s assets, liabilities, and income are included in the consolidated statements, but part of the equity belongs to outside shareholders.

Two equity concepts are therefore different:

- `StockholdersEquity` : equity attributable only to parent-company shareholders;
- `StockholdersEquityIncludingPortionAttributableToNoncontrollingInterest` : parent-company equity plus NCI.

If Alpha Vantage uses the latter in one year and the former in another year, the field name stays the same, but the scope of the time series has changed.

Accession Number

An accession number is the unique identifier assigned by EDGAR to an electronic submission, for example:

0000950170-24-075605

It points to a specific filing submission, not merely to a company or a fiscal year. Its format generally includes the filer CIK, the year, and a sequence number.

Why “Same Accession” Matters

“Same accession” means all components in a formula come from the same SEC submission. For example, it is not acceptable to take assets from a 2024 annual report, subtract equity restated in a 2025 annual report, and claim the result comes from one auditable 2024 filing. Mixing across accessions can combine different versions, different scopes, or different restatement states. Therefore, this study requires derived liabilities and debt reconciliations to use components from the **same period, same unit, and same accession**.

2.5 How the 294 Eligible Cells Are Calculated

An eligible cell is the smallest predefined research unit:

one company × one fiscal year × one core metric

It does not require Alpha Vantage or SEC to actually have a value. As long as the cell is within the research scope, it counts as eligible. This avoids silently shrinking the denominator because of missing data.

The calculation is:

7 companies × 6 fiscal years × 7 metrics = 294

The study then distinguishes several layers:

- **Endpoint available:** whether the Alpha Vantage endpoint succeeds;
- **SEC-verifiable:** whether the product definition can be supported by SEC evidence;
- **Comparable:** whether both the Alpha Vantage value and the SEC reference matching the product definition exist;
- **Flagged:** whether the cell requires a warning, review, or high-risk classification.

Among the 294 eligible cells, 290 have a complete SEC reference matching the product definition. The remaining four are: two CVX cash cells where an exact cash reference is missing, and Ford FY2019/FY2020 net income, where only income available to common shareholders was found in the checked evidence and no separate parent-net-income fact was found.

3. Audit Conclusions for the Seven Common Metrics

Metric	Sample result	Main risk	Final judgment
Total assets	42/42 match the product definition; 41 exact, 1 rounding difference	Very low	Best suited for the Alpha Vantage fast path
Total liabilities	41/42 match latest-known values; 21/42 require derivation	Missing direct tags; version selection	Usable, but direct/derived status must be disclosed
Parent	37/42 match; 5/42 actually match equity including NCI	Scope drift across companies	Requires company-year

equity		or years	reconciliation
Cash	Among 40 verifiable cells, 39 match and 1 uses an older version; 2 are unverifiable	Restricted cash and historical versioning	Exact tag can be fast path; otherwise do not guess
Revenue	30/42 match broad revenue; 11/42 match narrower revenue; 1/42 uses an older version	Scope and version drift are most concentrated here	Cannot rely only on <code>totalRevenue</code>
Parent net income	Among 40 verifiable cells, 39 match and 1 matches common-shareholder income; 2 only have a common-shareholder concept found	Parent versus common-shareholder income	Requires reconciliation
Operating cash flow	40/42 exact or rounding; 1 small difference; 1 large unexplained difference	Rounding and vendor mapping	Generally stable, but exceptions must be monitored

3.1 Total Assets

All 42 cells match the product definition. IBM FY2020 differs from SEC by 1 million dollars; relative to roughly 156 billion dollars of assets, this is a display-precision difference.

Assets is the most stable metric in this sample. But it should not be used as evidence that all standard metrics are equally stable, because revenue, equity, and net income involve scope choices.

3.2 Total Liabilities: Matching the Number Does Not Mean the Filing Has a Direct Tag

A concrete example is ORCL FY2024. In Company Facts for this period, there is no direct `Liabilities` fact. But within the same accession, `0000950170-24-075605`, the following components exist:

Assets	140.976B
– Equity including NCI	9.239B
= Derived total liabilities	131.737B

Alpha Vantage `totalLiabilities` is also 131.737B.

The conclusion is that the Alpha Vantage number can be reproduced by inference, but it is not a direct SEC fact. The product source panel should therefore say “Derived: Assets – equity including NCI” and show the accession numbers of the components. Otherwise, users will not find a directly disclosed balance-sheet line named 131.737B, and they may wrongly believe the source link is broken or inconsistent.

3.3 Shareholders’ Equity: IBM Shows That the Same Field Can Change Meaning Within One Company

IBM’s Alpha Vantage `totalShareholderEquity` behaves as follows:

IBM FY	AV	Parent equity	Equity including NCI	Actual AV scope
2019	20.985B	20.841B	20.985B	Includes NCI
2020	20.725B	20.597B	20.727B	Includes NCI, 2M rounding
2021	18.996B	18.901B	18.996B	Includes NCI
2022	21.944B	21.944B	22.021B	Parent
2023	22.533B	22.533B	22.613B	Parent
2024	27.307B	27.307B	27.393B	Parent

FY2022 is the switching point. The field name does not change. But FY2019–FY2021 include the equity held by third-party owners of consolidated subsidiaries, while FY2022 onward excludes it. That means a change in accounting scope is mixed into what appears to be a simple time-series trend.

CVX FY2021 and GM FY2024 also show Alpha Vantage matching equity including NCI rather than the parent-equity definition used in this study.

3.4 Cash: IBM FY2020 Shows Historical Version Risk

IBM FY2020:

- As-first-reported cash: 13.212B;
- Later latest-known cash: 13.188B;
- Current Alpha Vantage historical value: 13.212B.

Alpha Vantage matches the as-first-reported version, not the later version. The difference is 24M. This number looks normal; it does not appear missing or obviously broken. That makes it a **silent historical-version risk**.

Separately, CVX FY2019 and FY2024 lack the exact `CashAndCashEquivalentsAtCarryingValue` reference required by this study. Although the

filing may contain cash-plus-restricted-cash data, that cannot automatically be treated as ordinary cash. The correct status is unverifiable, not guessed decomposition.

3.5 Revenue: XOM and CVX Show That Field Meaning Can Drift Across Companies and Years

This study defines revenue as broad consolidated top-line revenue. For XOM, Alpha Vantage `totalRevenue` consistently matches the narrower “Sales and other operating revenue” line across all six years, rather than broad revenue:

XOM FY	AV / narrow sales	Broad revenue	Difference
2019	255.583B	264.938B	9.355B
2020	178.574B	181.502B	2.928B
2021	276.692B	285.640B	8.948B
2022	398.675B	413.680B	15.005B
2023	334.697B	344.582B	9.885B
2024	339.247B	349.585B	10.338B

These are not unexplained arithmetic errors. Alpha Vantage is using another visible financial-statement line.

CVX is more dangerous: FY2019–FY2021 use the narrow revenue line, FY2022 suddenly uses broad revenue, and FY2023–FY2024 return to narrow revenue. Even if a user only looks at a single company, Alpha Vantage’s `totalRevenue` time series is not semantically consistent across years.

IBM FY2020 also shows a version issue:

```
AV revenue:          73.621B
As-first-reported: 73.620B
Latest-known SEC:   55.179B
```

Alpha Vantage is closer to the as-first-reported version, while the latest-known value has been restated.

3.6 Net Income: Different Concepts Sometimes Match Numerically, but Sometimes Affect the Chart

GM FY2023:

```
AV netIncome          10.022B
NetIncomeLoss attributable to parent 10.127B
Available to common   10.022B
```

Alpha Vantage matches income available to common shareholders, not parent net income. The difference is 105M. The reason is that income available to common shareholders deducts items such as preferred-stock effects or participating securities before arriving at the amount available to ordinary common shareholders.

Ford FY2019 and FY2020 only have the available-to-common concept found in the checked evidence, and Alpha Vantage matches it. Because no separate parent-net-income fact was found, this does not prove the parent value is different. Therefore, Ford is a low-risk definition warning: the concept differs, but no observed numerical difference is confirmed. It should not be mixed with GM, where the concept difference creates an actual numerical difference.

3.7 Operating Cash Flow: Generally Stable, With One Large Unexplained Difference

Among 42 cells, 40 are exact matches or rounding differences. For CVX FY2019–FY2022, Alpha Vantage values appear to use 0.1B coarse rounding. For example:

```
CVX FY2020 AV operating cash flow: 10.600B
CVX FY2020 SEC operating cash flow: 10.577B
Difference:                        23M
```

This is more consistent with vendor rounding than with a separate economic definition.

IBM FY2023 is different:

```
AV operatingCashflow 13.432B
SEC operating cash flow 13.931B
Difference            0.499B, about 3.58%
```

Among the concepts, periods, and versions checked, no alternative SEC line was found that explains this difference. It remains the only large

unexplained pure-numerical difference retained by the study.

4. SEC As-First-Reported Values Versus Later Restatements

4.1 A Change in Representation Is Not the Same as a Change in Value

The original program timeline classified 24 cells as having changed in later presentation. But that set included several cases where the economic value did not truly change:

- The concept changed, but the value stayed the same;
- A direct fact became a derived representation;
- The scope or tag changed, but the number did not change.

After manual audit, only **six cells** show a true numerical change. The other 18 are changes in representation or concept only.

Company / FY / Metric	Earliest	Latest	Current AV	AV closer to
IBM 2019 revenue	77.147B	57.714B	57.714B	Latest
IBM 2020 revenue	73.620B	55.179B	73.621B	Earliest
IBM 2020 cash	13.212B	13.188B	13.212B	Earliest
GM 2022 liabilities, derived	191.752B	192.110B	191.753B	Earliest
GM 2023 liabilities, derived	204.757B	204.875B	204.757B	Earliest
IBM 2019 liabilities, derived	131.202B	131.201B	131.201B	Latest

Among these six true numerical-change cells:

- Alpha Vantage is closer to earliest: 4/6;
- Alpha Vantage matches latest: 2/6;
- Neither: 0/6.

The sample is too small to estimate Alpha Vantage's market-wide historical backfill policy. But it is sufficient to show that Alpha Vantage does not uniformly backfill all history to latest values, nor does it uniformly preserve as-first-reported values.

4.2 What Happened to IBM FY2019

IBM FY2019 revenue:

1. The FY2019 filing first disclosed 77.147B;
2. The FY2020 10-K still showed 77.147B;
3. The FY2021 10-K restated FY2019 revenue to 57.714B;
4. Later filings no longer re-presented FY2019 revenue;
5. Alpha Vantage's current FY2019 revenue is 57.714B, matching the latest-known value.

IBM FY2019 net income did not change among the later re-presentations observed. When later filings no longer re-present FY2019, the correct statement is “not re-presented later” — not “not modified later.” The absence of re-presentation is not proof that a value could never have been modified again.

5. Definition, Frequency, Size, Concentration, and Severity of Gaps

5.1 What “Silent Semantic or Historical-Version Difference” Means

In this report, a silent semantic or version difference must satisfy all of the following conditions:

1. Alpha Vantage returns a non-empty number of reasonable magnitude;
2. The number does not actively alert the user that something is wrong;
3. But the number does not correspond to the SEC concept, scope, or version required by the product definition;
4. A user clicking the product's claimed source filing or line cannot directly reproduce the number under the product definition;
5. The difference has a visible numerical effect.

Examples include:

- IBM equity includes NCI, while the product definition excludes NCI;
- CVX revenue uses broad revenue in one year and narrow revenue in other years;
- GM net income uses income available to common shareholders rather than parent net income;
- IBM FY2020 revenue uses an earlier historical version, while a later filing restated the value.

This is more dangerous than missing data, because the number looks normal.

5.2 What “Warning / Error Cells” Means

A flagged review cell is a research unit that requires a warning, explanation, review, or blocking. In this study, it means any eligible cell whose danger level is not `NONE`. After audit, there are **34** flagged review cells:

- **D1 — 2 visible gaps.** Example: CVX cash in certain years. Alpha Vantage has a cash value, but no exact SEC cash fact satisfying the product definition was found.
- **D2 — 11 low-impact definition or rounding warnings.** These are cases where the concept differs but the current number is the same, or the difference is small and does not currently affect the trend. Example: Ford FY2019 net income — Alpha Vantage matches income available to common shareholders, a 47M item. The study’s preferred SEC definition is parent net income, but no independent parent-net-income fact was found for that period to prove the value differs.
- **D3 — 1 large unexplained numerical difference.** Example: IBM FY2023 operating cash flow.
- **D4 — 20 silent semantic or version differences.** Examples:
 - IBM FY2019 equity: Alpha Vantage includes NCI, while the SEC product definition excludes it;
 - CVX FY2024 revenue: Alpha Vantage uses narrower customer/sales revenue, while the SEC definition is broad revenue;
 - GM FY2023 net income: Alpha Vantage uses available-to-common income of 10.022B, while parent net income is 10.127B;
 - IBM FY2020 revenue: Alpha Vantage uses an earlier historical version, while a later filing restated the value.

Therefore, the 34 flagged cells do not mean “34 Alpha Vantage errors.” They include possible vendor semantic choices, missing exact SEC references, reasonable rounding, historical-version choices, and truly unexplained numerical differences.

5.3 Frequency

Among the 294 eligible cells:

Status	Count
Alpha Vantage endpoint and field available	294/294
Product definition fully verifiable from SEC	290/294
Product contract matched	268/290 (92.4%)
Silent semantic or historical-version difference with numerical effect	20/290 (6.9%)
Numerical difference not yet explained by semantics or versioning	2/290 (0.7%)
Full product definition not verifiable	4/294

The two unexplained differences are CVX FY2020 operating cash flow (a 23M rounding-type difference) and IBM FY2023 operating cash flow (a large 499M difference).

5.4 Size

For the 20 silent semantic or version differences:

Statistic	Value
Median absolute difference	3.482B
p90 absolute difference	10.338B
Maximum absolute difference	18.442B (IBM FY2020 revenue version difference)
Median relative difference	about 2.44%
Maximum relative difference	about 33.42%

This shows that the main risk is not decimal-place noise. The main risk is that the product is using another financial-statement line or another historical version.

5.5 Concentration

Among the 34 flagged review cells, revenue contributes 13 cells; equity, net income, and operating cash flow each contribute 5 cells.

Among the 20 D4 silent semantic or version differences:

Metric	D4 count
Revenue	12/20
Equity	5/20
Net income	1/20
Cash	1/20
Liabilities	1/20

Revenue plus equity account for 17/20 D4 cases, or 85%. The problem is clearly concentrated in scope-sensitive metrics. It is not evenly distributed across all fields.

5.6 Severity Levels

Level	Name	Meaning
D1	Visible gap	SEC product reference is missing or clearly unverifiable; users can usually see the gap
D2	Low-impact definition or rounding warning	Concept differs but current value is the same, or the difference is only minor rounding
D3	Material unexplained numerical difference	Large numerical difference with no confirmed semantic or version explanation yet
D4	Silent semantic or version difference	Number exists and looks reasonable, but scope, concept, or version does not satisfy the product definition and has a visible numerical impact

Typical D4 cases include: IBM equity switching across years between including and excluding NCI; CVX revenue reversing scope in FY2022; XOM using narrow sales for all six years; GM FY2023 common-shareholder income versus parent net income; and IBM FY2020 revenue and cash using as-first-reported values.

6. Are Same-Industry, Same-Name Metrics Comparable?

6.1 MSFT / ORCL (SIC 7372)

The core financial metrics are generally the cleanest in this pair. ORCL liabilities often need to be derived using the balance-sheet identity; therefore, the values are comparable, but the source presentation is not the same. Charts or API metadata should mark whether the value is direct or derived.

Revenue may use different exact SEC concepts, so one must verify whether both companies' values still represent consolidated top-line revenue. RPO and cloud revenue should not be plotted together merely because the names sound similar. For example, Microsoft discloses cloud revenue as an absolute amount, while Oracle discloses cloud revenue in some periods as a percentage of total revenue. The unit and expression differ, so they are not directly comparable without transformation and clear labeling.

6.2 XOM / CVX (SIC 2911)

Balance-sheet metrics are mostly comparable. Revenue should not be used directly from Alpha Vantage `totalRevenue`: XOM uses narrow sales for all six years, while CVX uses narrow revenue in five years and broad revenue in one year.

Production volume can be converted into boe/day for comparison, but further checks are still needed for net versus gross production, equity-method scope, and product mix. Proved reserves can also be standardized by unit, but footnotes are still needed for geographic scope, equity basis, and oil-versus-gas composition.

6.3 F / GM (SIC 3711)

Core metrics are generally comparable, but two issues remain: GM FY2023 net income has a scope issue, and GM FY2024 equity has a scope issue. Ford FY2019 and FY2020 do not have an independent parent-net-income fact found in the checked evidence; income available to common shareholders can only be treated as substitute evidence, not as proof that the parent-net-income value is identical.

Vehicle-related KPIs are not automatically comparable: wholesales, retail sales, deliveries, and total vehicle sales are not natural synonyms. Auto finance subsidiaries also affect the consolidated scope of assets, liabilities, revenue, and debt.

7. Industry-Specific and Company-Specific KPIs: How Candidates Were Generated and Whether Alpha Vantage Covers Them

7.1 Were These KPIs Automatically Suggested by SEC?

No. This study uses **hypothesis-driven targeted exploration**. It does not ask SEC to automatically generate a full KPI list.

The candidate KPIs come from two sources:

1. The original task requirement to explore industry-specific or company-specific KPIs;
2. Researcher-defined candidate families based on business models: - Software: RPO, cloud revenue, remaining performance obligations; - Oil: production volume, proved reserves; - Automotive: wholesales, retail sales, deliveries; - IBM: RPO.

The study then checks SEC filings to verify whether these KPIs actually exist, where they are disclosed, and whether they are structured.

RPO has the standard concept `us-gaap:RevenueRemainingPerformanceObligation`, so it can be found directly in Company Facts. Cloud revenue, production volume, reserves, and vehicle sales are mainly found through tables, text searches, and company-specific extraction rules in the latest two 10-K filings.

Therefore, the 13 KPI records represent validation of seven preselected KPI families across the relevant companies. They are not an exhaustive enumeration of all special KPIs in the filings. The report should describe this as **targeted exploration**, not as a full discovery of all company-specific KPIs.

7.2 Actual Findings

Among 13 KPI exploration records:

Result	Count
Quantitative SEC disclosure found	12/13
Comparable SEC total not found	1/13 (Ford retail-sales total)
Same-semantics field found in Alpha Vantage's four fundamentals endpoints	0/12
Meaningful proxy found in Alpha Vantage's four fundamentals endpoints	0/12

SEC disclosure structure:

Disclosure structure	Count
Standard XBRL	3 (MSFT / ORCL / IBM RPO)
Filing tables	6
Filing text	3
Not found	1

Examples of the disclosed values:

- MSFT RPO: 375B; ORCL RPO: 638B; IBM RPO: 71B;
- MSFT cloud revenue: 168.9B; ORCL cloud revenue: disclosed as 51% of total revenue;
- XOM production: 4.736M boe/day; XOM proved reserves: 19.311B boe;
- CVX production: 3.7M barrels/day; CVX reserves: 10.591B boe;
- Ford wholesales: 4.395M units; GM wholesales: 3.799M units;
- GM total vehicle sales: 6.184M units — including retail, fleet, and certain dealer-use vehicles, which is not the same as Ford retail sales.

A generic field such as `RevenueTTM` is not a proxy for cloud revenue, production volume, reserves, or vehicle sales: the business object, unit, and analytical purpose are different.

Conclusion: within the Alpha Vantage fundamentals endpoints tested in this study, same-semantics coverage for the targeted special KPIs is **0/12**. This conclusion should not be generalized to all Alpha Vantage product lines, but it is enough to show that the four basic fundamentals endpoints do not satisfy this requirement.

8. Total Debt as a Derived Metric: What Comes from SEC Facts and What Is Researcher Inference

8.1 SEC Does Not Provide a Universal Formula for “Total Debt”

SEC provides standard concepts, their definitions, and facts reported by companies. Examples include:

Concept	Meaning
DebtCurrent	Current total of short-term debt, current maturities of long-term debt, and certain lease obligations
LongTermDebt	Carrying amount of long-term debt; taxonomy definition excludes capital / finance leases
LongTermDebtNoncurrent	Noncurrent portion of long-term debt
LongTermDebtAndCapitalLeaseObligations	Parent concept combining noncurrent long-term debt and lease obligations
FinanceLeaseLiability	Present value of finance-lease payment obligations
OperatingLeaseLiability	Present value of operating-lease payment obligations
DebtLongtermAndShorttermCombinedAmount	Aggregate of long-term debt (including current maturities) and short-term debt

These concept names, definitions, and fact values come from SEC / XBRL.

8.2 What This Study Defines and Infers

The following are **researcher-defined** product scope and resolution paths, not an SEC-official universal total-debt formula:

1. Whether the product metric includes finance leases or excludes them;
2. Which parent/component combinations to try when a direct combined-debt tag is missing;
3. When it is acceptable to subtract finance leases from an inclusive debt concept;
4. When the result should be marked as candidate or partial only;
5. Whether same-value facts should trigger an overlap red flag;
6. The requirement that all additions and subtractions use the same accession, same period, and same unit.

In other words: SEC provides the raw materials and semantic definitions; this study defines the metric contract and derivation assumptions, then tests them against SEC facts through falsification and reconciliation.

8.3 Why the Study Uses Broad Current Debt and the Debt-and-Capital-Lease Parent Path

These paths are not arbitrary guesses. They are candidate identities built from the explicit definitions of taxonomy concepts. For example:

```
DebtCurrent
+ LongTermDebtAndCapitalLeaseObligations
```

This attempts to cover current debt plus noncurrent debt-and-lease. If the product metric is intended to exclude finance leases, then a separately identified finance-lease liability may need to be subtracted from the same accession.

But the formula is acceptable only when all of the following conditions hold:

- Parent/component scope is clear;
- Components do not overlap;
- Components come from the same accession;
- Period and unit are the same;
- Finance-lease liability can be separately identified;
- The arithmetic reconciliation holds.

If these conditions are not met, the result must be returned as candidate, partial, or unresolved.

8.4 Why These Derivations Are Not Ground Truth

This study did not manually read every debt footnote for every debt case to confirm issuer-specific custom tags, finance-subsidary dimensions, measurement basis, and the economic substance of all debt instruments. Therefore, “fully resolved” is more accurately described as:

Resolved under standard-tag semantics and same-accession reconciliation.

It is not an absolute certification of the full economic substance of every filing.

Among 24 debt metric rows:

Resolution status	Count
Fully resolved	8

Candidate	4
Partial	6
Unresolved	6

This proves that a fixed three-tag formula is not universal. But it does not mean SEC data can never be used to construct debt metrics. The correct approach is to layer the evidence by metric contract, tag family, and confidence level, and to abstain when evidence is insufficient.

9. Final Conclusions

1. Alpha Vantage is not broadly unusable. Simple headline facts such as total assets perform well in this sample.
2. The main risk is not frequent arithmetic error. The main risk is unstable semantics, scope, or historical versioning under the same field name.
3. IBM equity and CVX revenue prove that the same Alpha Vantage field can change meaning across years within the same company.
4. Among 290 fully comparable cells, 268 match the product definition; 20 have numerically visible silent semantic or version differences; and 2 have numerical differences that are not yet fully explained.
5. Alpha Vantage's historical backfill behavior is not uniform: among six cells with true numerical changes, four are closer to as-first-reported SEC values and two match latest-known values.
6. In the targeted SEC special-KPI exploration, same-semantic coverage by the four Alpha Vantage fundamentals endpoints is 0/12.
7. Peer comparability must be based on metric contracts and SEC scope, not on API field names.
8. The total-debt formula is a researcher-defined candidate derivation tested against SEC facts. It is not an SEC-official universal definition. When evidence is insufficient, the product must abstain rather than force a number.